WORKED SOLUTION

STEP II 2007, Question 6

Pure • First-Order Differential Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q6

The key idea

Part (i) hands us two functions whose derivatives both have $\sqrt{3+x^2}$ in the denominator; a suitable combination of them has derivative a multiple of $\sqrt{3+x^2}$ itself, so the integral comes for free. In part (ii) the differential equation is a quadratic in $\frac{dy}{dx}$. Solving that quadratic gives two ordinary first-order equations, and each one needs exactly the integral from part (i).

Solution

Part (i)

Write $s = \sqrt{3+x^2}$, so that $s > 0$ for every real x and $\dfrac{ds}{dx} = \dfrac{x}{s}$. Note also that $x + s > 0$, since $s > |x|$, so the logarithm is defined for all x.

By the chain rule,

$$\frac{d}{dx}\ln(x+s) = \frac{1+\frac{x}{s}}{x+s} = \frac{\frac{s+x}{s}}{x+s} = \frac{1}{\sqrt{3+x^2}}.$$

By the product rule,

$$\frac{d}{dx}(xs) = s + x\cdot\frac{x}{s} = \frac{s^2+x^2}{s} = \frac{3+2x^2}{\sqrt{3+x^2}}.$$

? **Ask yourself** How do these two derivatives give $\int \sqrt{3+x^2}\,dx$?

We want a numerator of $3+x^2$ over the denominator s, since $\dfrac{3+x^2}{s} = s$. The second

derivative has numerator $3+2x^2$; adding 3 times the first derivative gives

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(xs+3\ln(x+s)\Big)=\frac{3+2x^2+3}{s}=\frac{2(3+x^2)}{s}=2\sqrt{3+x^2}.$$

Therefore

$$\int\sqrt{3+x^2}\,\mathrm{d}x=\tfrac{1}{2}x\sqrt{3+x^2}+\tfrac{3}{2}\ln(x+\sqrt{3+x^2}\,)+C.$$

Part (ii)

Write $p=\dfrac{\mathrm{d}y}{\mathrm{d}x}$. The equation is the quadratic $3p^2+2xp-1=0$ in p, so

$$p=\frac{-2x\pm\sqrt{4x^2+12}}{6}=\frac{-x\pm\sqrt{3+x^2}}{3}.$$

The discriminant $4x^2+12$ is always positive, so there are two different real roots at every x. Their difference, $\frac{2}{3}\sqrt{3+x^2}$, is never zero, so a solution can't switch from one root to the other part-way along. (A derivative has the intermediate value property, so it can't jump between two values that are always at least $\frac{2}{3}\sqrt{3}$ apart.) So every solution satisfies one of

$$\frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{x}{3}+\frac{1}{3}\sqrt{3+x^2}\qquad\text{or}\qquad\frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{x}{3}-\frac{1}{3}\sqrt{3+x^2}$$

for all x. Integrating with part (i),

$$y=-\frac{x^2}{6}\pm\Big(\tfrac{1}{6}x\sqrt{3+x^2}+\tfrac{1}{2}\ln(x+\sqrt{3+x^2}\,)\Big)+C.$$

At $x=1$ we have $\sqrt{3+x^2}=2$ and $\ln(x+\sqrt{3+x^2})=\ln 3$.

Taking the + sign: $y(1)=0$ gives $0=-\frac{1}{6}+\frac{1}{3}+\frac{1}{2}\ln 3+C$, so $C=-\frac{1}{6}-\frac{1}{2}\ln 3$. Combining the logarithms,

$$y=\frac{x\sqrt{3+x^2}-x^2-1}{6}+\frac{1}{2}\ln\left(\frac{x+\sqrt{3+x^2}}{3}\right).$$

Taking the − sign: $y(1)=0$ gives $0=-\frac{1}{6}-\frac{1}{3}-\frac{1}{2}\ln 3+C$, so $C=\frac{1}{2}+\frac{1}{2}\ln 3$, and

$$y=\frac{3-x^2-x\sqrt{3+x^2}}{6}-\frac{1}{2}\ln\left(\frac{x+\sqrt{3+x^2}}{3}\right).$$

These are the two solutions through $(1,0)$, and both are valid for all real x.

Checks

- Put $x=1$ in each answer: the bracket is $\frac{2-1-1}{6}=0$ (or $\frac{3-1-2}{6}=0$), and $\ln\frac{1+2}{3}=0$. So $y(1)=0$ in both.

- The two roots for p add up to $-\frac{2x}{3}$, so the two solutions should add up to $\frac{1-x^2}{3}$. The square roots and logarithms do cancel, leaving exactly $\frac{2-2x^2}{6} = \frac{1-x^2}{3}$.
- Numerically, Simpson's rule gives $\int_{-1}^{2} \sqrt{3+x^2}\,\mathrm{d}x \approx 5.949681$, matching the formula in part (i). Stepping each branch of the equation from $(1, 0)$ to $x = 4$ by Runge–Kutta gives $y \approx 0.584957$ and $y \approx -5.584957$, which match the two formulae. (Their sum is $-5 = \frac{1-16}{3}$.)

Common slips

- Treating $3p^2 + 2xp = 1$ as linear, or trying to separate variables, instead of solving it as a quadratic in p.
- Using one constant C for both branches. Each sign needs its own constant, fixed by $y(1) = 0$.
- Forgetting the factor $\frac{1}{3}$ in front of $\sqrt{3+x^2}$, so the part (i) integral is used with the wrong multiple.

Beyond STEP

An equation like $F\left(x, y, \frac{\mathrm{d}y}{\mathrm{d}x}\right) = 0$ that is not linear in the derivative is *implicit*. Through a point there can be several solutions, one for each root of the equation in p. Where two roots meet, *singular solutions* can appear (the envelope of a family of curves, as in Clairaut's equation). Here the roots never meet, so there are exactly two solutions. The $\ln\left(x + \sqrt{3+x^2}\right)$ in part (i) is $\operatorname{arsinh}(x/\sqrt{3})$ plus a constant, and the integral is the standard $x = \sqrt{3}\sinh u$ substitution in disguise. Boyce and DiPrima, *Elementary Differential Equations*, covers first-order equations and existence and uniqueness of their solutions.

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